SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

FIN207

Examination - January Semester 2024

Game Theory and Design Thinking

Friday, 26 April 2024 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **5** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Section A (100 marks)
Answer all questions in this section.

Question 1

There is a simultaneous game between Bill and Kelly. The game table is as follows.

		Kelly	
		Yes	No
Bill	Yes	-5, -6	-4, 5
	No	5, -4	-6, -5

Question 1a

Analyse and find the pure-strategy Nash Equilibrium(s) or explain why if there is none.

(10 marks)

Question 1b

Indicate which of the following games is similar to the above game and explain your choice.

1. Assurance game
2. Pure coordination game
3. Battle of sexes
4. Chicken game
5. None of the above choices

(10 marks)

Question 1c

Assume that you are one of the player Kelly. Indicate which strategy move (i.e., Threat, Promise, or Commitment) that you can use when playing the game to obtain a better outcome and explain your choice.

(10 marks)

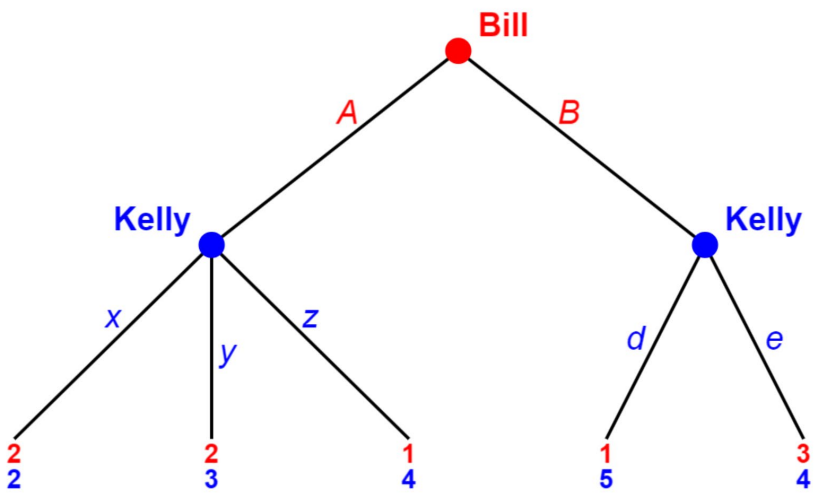
Question 1d

Justify if it is possible to have any mixed strategy equilibrium or explain why there is none.

(10 marks)

Question 2

Consider a sequential game between Bill and Kelly as follows.



Question 2a

Show each player's available strategies.

(8 marks)

Question 2b

Find the Rollback equilibrium(s) or explain why if there is none.

(12 marks)

Question 3

Bill and Kelly are two milk tea shop owners in CCK mall. They are allowed to update the price information for their public display E-board on the first day of each month. Initially they agreed to set the same price for one identical milk tea, and this agreement is believed to continue forever.

Consider the monthly repeated updating process as a repeated simultaneous game between Bill and Kelly. Both can choose to cooperate to maintain the agreement or not to cooperate on the date of updating the price. The payoff table is given as follows.

		Kelly	
		Cooperate	Not Cooperate
Bill	Cooperate	10,8	0,9
	Not Cooperate	16,0	4,6

Question 3a

Verify that the game is a prisoners' dilemma.

(10 marks)

Question 3b

Assume both players use a Grim strategy. Justify if it is worthwhile for Bill to defect just once, given the rate of return r=0.1.

(10 marks)

Question 4

Suppose you have a project in the newly developed AI area. If it succeeds, you are expected to make a profit of $520,000. If it fails, it will be 0 profit. Now you plan to hire a manager to manage the entire project. Assume E is the effort level with a value of 1 or 0. If the manager exerts full effort (E=1) on the project, the success probability of the project is 60%. If the manager exerts 0 effort (E=0) on the project, the success probability is 30%.

Let P represent the payment to the manager. The manager's utility function is U=P-1000*E. Assume the manager can work for other jobs without any effort to get a payment of $10,000.

Question 4a

Assume that the effort level is observable, calculate the payment to the manager if you require no effort and the payment to the manager if you require full effort.

(10 marks)

Question 4b

Assume the effort level is unobservable. You would like to provide the incentive to the manager by linking the payment with the outcome of the project. You will pay S to the manager if the project succeeds and F to the manager if the project fails.

Present the expression of the expected utility (by using S and F) if the manger exerts full effort and present the incentive compatibility constraint to motivate full effort.

(10 marks)

----- END OF PAPER -----